

Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.

Team Members	Role/ Responsibility	Email				
Constantine Akas	EM Algorithm Implementation, Data Exploration, GitHub Management	c.akas@alustudent.com				
Ndunge Mutheu Mbithi	Bayesian Probability Analysis	n.mbithi@alustudent.com				
Regina Anthony Majura	Manual Gradient Descent Calculations	r.majura@alustudent.com				
Keyla Nyacyesa Bineza	Gradient Descent Coding & Visualization	k.nyacyesa@alustudent.com				

Task Allocation and Tracker

	Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments
	Download and explore Galton Heights dataset	Constantine Akas	June 27	Completed	Yes	Dataset loaded and analyzed
	Create mixed height distribution	Constantine Akas	June 27	Completed	Yes	Father and child heights combined
	Histogram visualization	Constantine Akas	June 28	Completed	Yes	Distribution visualized
	Implement E-Step	Constantine Akas	June 28	Completed	Yes	Responsibilities calculated
	Implement M-Step	Constantine Akas	June 29	Completed	Yes	Parameters updated
	Log-likelihood calculation	Constantine Akas	June 29	Completed	Yes	Working successfully
	EM tracking table (Iterations 0–2)	Constantine Akas	June 29	Completed	Yes	Required rubric item completed
	Gradient Descent manual calculation	Regina Anthony Majura	July 1	Completed	Yes	the calculations were done and pushed on github as pdfs like requested
	Gradient descent code implementation	Keyla Nyacyesa Bineza	July 3	Completed	Yes	Implemented MSE, gradients, SciPy derivative functions
	Gradient descent code implementation	Keyla Nyacyesa Bineza	July 5	Completed	Yes	Implemented gradient descent plots
	Bayesian Probability	Ndunge Mbithi	July 4	Completed	Yes	prior probability calculation
	Bayesian Probability	Ndunge Mbithi	July 4	Completed	Yes	likelihood calculation
	Bayesian Probability	Ndunge Mbithi	July 5	Completed	Yes	marginal and posterior probability calculation

Meeting Attendance Log						
Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps	
June 29, 2026	Initial project setup and Part 1 progress update	Constantine	Regina Anthony Majura , Keyla Bineza Nyacyesa,Ndunge Mutheu Mbithi	Explored dataset, implemented EM algorithm, created tracking table	Share notebook with team and assign remaining tasks	
July 1 2026	feed back	Constantine	Regina Anthony Majura , Keyla Bineza Nyacyesa,Ndunge Mutheu Mbithi	see the progress of the work	members had go start pushing progress to github	
[Date]		[Name]	[List of Attendees]	[Summary of Discussion]	[Follow-up Actions]	
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